

value of the floor will increase as rates go down, whereas the value of a 1% annuity won't change too much when rates move, and thus it is clear that this option will be exercised in a low interest rate environment. This option is a compound option, as it gives the holder an option to buy a floor, which is an option itself. Valuing this compound option itself is beyond the scope of this discussion.²⁵

²⁵As mentioned, this is a trade that was proposed to a dealer on reverse inquiry. The client wanted a price on the option back in 20 minutes. After 5 minutes, the group at the dealer that was trying to price the option was still not even certain in what type of interest rate environment the option would be exercised, much less the price of the option. But one person in the group realized that this question could be answered by invoking cap-floor parity, as indicated above.

Problem Set 4

For questions 1–3, consider the following swap curve:

Maturity	Swap Rate
1 Year	5.60%
2 Year	6.00%
3 Year	6.20%
4 Year	6.40%
5 Year	6.60%

These rates are annual bond versus 1-year LIBOR paid annual bond as well.

In addition, for questions 1–3, consider the following base vol curve for 1-year LIBOR forwards:

Forward Period	Base Vol
0x1	n.a.
1x2	13%
2x3	14%
3x4	16%
4x5	15%

1. A client of yours is interested in purchasing a cap on 1-year LIBOR, struck at 6.50%, with a notional of \$100mm. The maturity of the cap is five years.

a) What is the value of the cap (expressed in both dollars and bps of the notional)?

b) Use the value that you computed in part a) to determine the average vol of the cap. That is, if you constrain each of the base vols to be the same number, what vol level provides a value for the cap that is equal to the value in part a)?

c) What is the value of a 5-year floor on 1-year LIBOR also struck at 6.50%? In order to price the floor, use the average vol from part b). Be sure to include the floorlet that resets today and pays in one year. There is no optionality to this floorlet, but there is a known payment that is to be included.

d) Using the values of the cap and floor computed above, use cap-floor parity to determine the value of a 5-year swap in which a counterparty pays a fixed rate of 6.50%. Use the bootstrap method to verify the value of the swap with a 6.50% fixed rate.

e) Now your client is interested in a costless collar. Your client wants to buy a 5-year cap struck at 7.00% and sell to you a 5-year floor at some strike rate x such that the value of the net position is zero. Price both the cap and the floor using the average vol from part b). What is the strike rate x on the floor? When pricing the floor, be sure to include the floorlet that pays in one year.

f) You check with your trader before trading the costless collar. She agrees with your methodology for determining the strike on the floor in part e) but indicates that this is a mid-market strike rate. She says that she is willing to buy or sell any cap or floor one vega from mid. By this she means that the average vol used to price a cap or a floor should be increased or decreased by one percentage point, depending on whether or not she is buying or selling the option. What is the strike on the floor in a costless collar, taking into account these bid/offer costs? What is the total profitability in dollars to you/your trader (profitability here means the difference in value between an option priced using a bid or offered side average vol and using a mid-market average vol)?

2. A client is interested in paying you fixed in a \$100mm 5-year swap and wants to reduce the fixed rate paid to you. You suggest that you pay 1-year LIBOR, capped at 8%. That is, you will pay $\min(\text{LIBOR}, 8\%)$ in the swap. What is the lowest swap rate you are willing to receive if you pay LIBOR capped at 8%? That is, what fixed rate makes this an $\text{NPV} = 0$ transaction?

3. Recall the convertible cap trade discussed in Section 4.1.7. Suppose that a dealer and a client have previously entered into the following trade: the dealer sells a 5-year cap on 1-year LIBOR struck at $X\%$ to the client for an upfront premium. This cap is like a regular cap except that one year after the effective date of the trade, the dealer has the right to

- (1) Call the cap from the counterparty *and*
- (2) Enter into a 4-year swap with the client in which the dealer receives $Y\%$ fixed annual bond and pays 1-year LIBOR.

Note that if the dealer calls the cap, then it must also enter into the swap. Suppose it is now one year after the effective date of the trade. The dealer

now has the right to exercise its option to call the cap away from the client and enter into a 4-year swap in which the dealer is receiving fixed at $Y\%$ annual bond versus 1-year LIBOR.

a) If $X = 7\%$ and $Y = 6\%$, should the dealer exercise its option and call the cap and enter into the swap?

b) If $X = 6\%$ and $Y = 4\%$, should the dealer exercise its option and call the cap and enter into the swap?

c) For this part, forget parts a) and b) above. Suppose that a dealer and a client have previously entered into the following trade: the dealer sells a 5-year floor on 1-year LIBOR struck at 6% to the client for an upfront premium. This floor is like a regular floor except that one year after the effective date of the trade, the dealer has the right to

- (1) Call the floor from the counterparty *and*
- (2) Enter into a 4-year swap with the client in which the dealer pays 8% fixed annual bond and the client pays 1-year LIBOR.

Note that if the dealer calls the floor, then she must also enter into the swap. Suppose it is now one year after the effective date of the trade. The dealer now has the right to exercise its option to call the floor away from the client and enter into a 4-year swap in which the dealer is paying fixed at 8% annual bond versus 1-year LIBOR.

Should the dealer exercise its option and call the floor and enter into the swap?

4. Recall the chooser cap discussed in Section 4.1.7. Consider again our stylized example and suppose that the swap curve is as follows:

Maturity	Swap Rate
1 Year	6.00%
2 Year	5.90%
3 Year	5.80%
4 Year	5.80%
5 Year	5.60%

These rates are annual bond versus 1-year LIBOR paid annual bond as well.

In addition, suppose the base vol curve is as follows:

Forward Period	Base Vol
0x1	n.a.
1x2	15%
2x3	18%
3x4	21%
4x5	20%

Suppose that sometime ago we purchased a chooser cap on 1-year LIBOR and that there is a LIBOR setting today for one of the caplets in the trade. Suppose further that we can choose to select a payment from just one more of the caplets remaining in the trade. That is, we have previously exhausted all but one of our choices in the chooser cap.

For each question below, specify which of the following applies:

- We should choose the caplet that sets today, thereby receiving a payment in a year.
- We should not choose the caplet that sets today but rather save our choice for the future.
- Without a more sophisticated model, it is not possible to determine what to do.

a) The chooser cap expires in two years. That is, there is the caplet that has a rate setting today and one additional caplet remaining. We can choose to either receive a payment in a year based on the rate setting today or save our choice. The strike on the chooser cap is 5.50%.

b) The chooser cap expires in two years. That is, there is the caplet that has a rate setting today and one additional caplet remaining. We can choose to either receive a payment in a year based on the rate setting today or save our choice. The strike on the chooser cap is 5.75%.

c) The chooser cap expires in three years. That is, there is the caplet that has a rate setting today and two additional caplets remaining. We can choose to either receive a payment in a year based on the rate setting today or save our choice. The strike on the chooser cap is 5.75%.

d) The chooser cap expires in three years. That is, there is the caplet that has a rate setting today and two additional caplets remaining. We can choose to either receive a payment in a year based on the rate setting today or save our choice. The strike on the chooser cap is 5.50%.

e) The chooser cap expires in three years. That is, there is the caplet that has a rate setting today and two additional caplets remaining. We can choose to either receive a payment in a year based on the rate setting today or save our choice. The strike on the chooser cap is 5.42%.

5. For this entire question, do not use a swap model or an option pricing model to obtain your answers. Use only handcheck methods to compute your answers. That is, you may use a financial calculator but no spreadsheet software.

Swaps 'R Us is a prestigious Wall Street investment bank that is a big player in the swaps market. One of the reasons that *Swaps 'R Us* is such a big player is that it is willing to do all transactions at mid-market. As a service to its many clients, *Swaps 'R Us* routinely makes available quotes on some frequently traded swap products.

The first set of quotes contains the swap curve. These are the fixed annual bond rates that *Swaps 'R Us* is willing to pay or receive versus 1-year LIBOR (paid annual bond as well). In addition, the zero coupon rates and 1-year LIBOR forwards are also provided:

Maturity	Swap Rate	Zero Coupon Rate	LIBOR Forwards
1	6.00%	6.00%	6.00%
2	6.25%	6.26%	6.52%
3	6.50%	6.52%	7.05%
4	6.75%	6.79%	7.61%

The next set of quotes are prices of caps and floors on 1-year LIBOR. These prices are based on the swap curve given above and an assumed constant base vol curve of 15%. *Swaps 'R Us* provides quotes on 6%, 7%, and 8% caps and 4%, 5%, and 6% floors (quotes based on a notional of \$1mm). To provide its customers with the greatest amount of information, *Swaps 'R Us* shows prices for each individual caplet and floorlet:

Maturity	6% Caplet	7% Caplet	8% Caplet
1	0	0	0
2	6,088	1,836	377
3	10,171	5,131	2,238
4	13,715	8,465	4,830
Sum	29,974	15,432	7,446

Maturity	4% Floorlet	5% Floorlet	6% Floorlet
1	0	0	0
2	1	118	1,516
3	11	229	1,462
4	24	272	1,308
Sum	36	619	4,285

Customers can buy or sell any of the caplets or floorlets at the prices shown above. Thus, for example, the price of a 3-year 7% cap on a notional of \$1mm is equal to $\$0 + \$1,836 + \$5,131 = \$6,967$, or 69.67 bps of the notional.

You work for an insurance company, and your boss has asked you to work up a few quotes for him, based on the prices made available by Swaps 'R Us.

a) What fixed rate will Swaps 'R Us be willing to pay in a 4-year swap if the insurance company pays 1-year LIBOR – 100 bps?²⁶

²⁶The astute reader will note that the quantity 1-year LIBOR – 100 bps received by Swaps 'R Us would be a negative amount if 1-year LIBOR were to ever set below 1%. We do not need to worry about embedding some type of option in order to avoid the coupons from being negative should this ever occur. The *Negative Interest Rate Method* defined by ISDA applies to cases in which a floating payment in a swap is a negative quantity. Under this method, a counterparty entitled to receive a floating payment that happens to be a negative quantity would actually *make* a payment equal to the absolute value of the negative amount. That is, in cases in which the quantity 1-year LIBOR – 100 bps is negative, the insurance company would actually receive the absolute value of 1-year LIBOR – 100 bps in addition to the fixed rate it receives. In other words, the insurance company would actually be receiving twice! Instead of utilizing the Negative Interest Rate Method, counterparties can alternatively agree to employ the *Zero Interest Rate Method*. Under this alternative, a floating payment in a swap is deemed to be zero when it would otherwise be negative. Again, we can feel free to ignore this issue for our purposes here.

- b) What fixed rate will Swaps 'R Us be willing to pay in a 4-year swap if the insurance company pays the minimum of 1-year LIBOR – 100 bps and 6%?
 - c) What fixed rate will Swaps 'R Us be willing to pay in a 4-year swap if the insurance company pays the maximum of 1-year LIBOR – 100 bps and 4%?
 - d) What is the 3x4 swap rate, that is, the fixed rate Swaps 'R Us is willing to pay or receive versus 1-year LIBOR in a 1-year swap, three years forward?
 - e) Consider a 3x4 swap in which Swaps 'R Us pays fixed and the insurance company pays the minimum of 1-year LIBOR – 100 bps and 6%. What is the fixed rate in this swap?
6. Referring to Section 4.1.3, what is the value of a 5-year floor on 3-month LIBOR struck at 3%?

Chapter 5

Swaptions

The second type of option frequently traded in the swaps market is swaptions. Swaptions are options on swaps, giving the holder the right, but not the obligation, to enter into a swap. We will discuss how swaptions are used by investors and how they are priced using Black's formula. We will discuss some of the shortcomings of Black's formula for the pricing of swaptions (and for the pricing of other interest rate options, for that matter) and introduce an alternative known as the normal model, which is now widely used throughout the industry. We will also discuss Bermudan swaptions.

5.1 An Introduction to Swaptions

A *swaption* is an option to enter into a swap. There are two categories of swaptions. A *receiver swaption* (sometimes called a *call swaption*) gives the holder the right to receive fixed (and pay floating) in a swap. A *payer swaption* (sometimes called a *put swaption*) gives the holder the right to pay fixed (and receive floating) in a swap. The words “receiver” and “payer” therefore refer to the position of the option holder as it pertains to the fixed leg of the underlying swap in the event of exercise.

The exercise of swaptions can either be *single European*, often simply called *European* (exercisable on only one date); *Bermudan* or *multi-European* (exercisable on a sequence of discrete dates); or *American* (exercisable anytime during a specified period of time). In this section, we will consider only European swaptions. Upon exercise, swaptions can be settled in one of two

ways. If a swaption is *physically settled*, then the two counterparties will actually enter into the swap. If the swaption is *cash settled*, then instead of actually entering into the swap, the value of the swap at the time of exercise will be determined, and the seller will pay that amount to the buyer.¹

As an example, consider a “1 into 4” 7% receiver swaption, semi bond versus 3-month LIBOR quarterly money. The holder of this option has the right to enter into a 4-year swap, starting in one year.² If the holder exercises, then the counterparty who is long the option will receive fixed at 7% semi 30/360 and will pay 3-month LIBOR quarterly Act/360. Upon exercise, the first fixed payment will occur 1.5 years from today (i.e., six months after expiration), and the final fixed payment will occur five years from today. Upon exercise, the first floating payment will occur 1.25 years from today (i.e., three months after expiration), and the final floating payment will occur five years from today. Assuming this swaption is European, the holder has the right to exercise in one year.³ If the swaption were Bermudan, then the holder would have the right to exercise in one year and, say, every six months thereafter.

In order to build some intuition, try answering the following two questions:

¹The process of determining the value of the swap is in accordance with our discussion of the marking-to-market of swaps in Chapter 3. The market standard is that swaptions are cash settled. In this way, counterparties avoid credit exposure to one another (and the need to post collateral to one another under the terms of a CSA) beyond the expiry of the option. If swaptions are physically settled, then upon exercise counterparties continue to have credit exposure to one another and may have to post collateral under the terms of a CSA.

²The “ n into m ” swaption-speak is easily confused with the “ n by m ” cap/floor-speak. When speaking of swaptions, an n into m swaption refers to an option that expires in n years and is exercisable into a swap that is m years long. Alternatively, an n by m (also written $n \times m$) cap refers to a cap whose first reset is in n years and whose final payment is in m years from today.

For example, consider both a 5 into 10 swaption and a 5 by 10 cap. The 5 into 10 swaption is an option that expires 5 years from today. Upon exercise, the two counterparties will enter into a 10-year swap (which therefore has a final maturity that is 15 years from today). On the other hand, the 5x10 cap has its first reset in 5 years and its final payment in 10 years.

³The standard in the market is that the buyer will provide the seller two New York and London Business Days notification prior to the effective date of the underlying swap if he wants to exercise the swaption. For example, suppose that a counterparty purchased a swaption on Tuesday, August 3, 2004. Recall that a spot starting swap would have an effective date $T + 2$ New York and London Business Days after the trade date, which in this case would be August 5, 2004. If a counterparty were to purchase a 1 into 4 receiver swaption, the effective date on the underlying swap would be Friday, August 5, 2005, and the buyer would be required to notify the seller of its intention to exercise the swaption on Wednesday, August 3, 2005.